Spring Day 09 notes

|- Setter and constructor Injection

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Maven Directory

```
FirstApp [Downloads master]
  src/main/java
    com.day09
      Resume.java
      Student.java
      application-context.xml
    SpringCore.FirstApp
  src/test/java
    SpringCore.FirstApp
      AppTest.java
  JRE System Library [JavaSE-1.8]
  Maven Dependencies
  src
  target
  pom.xml
```

xml file

dependecies

```java
package com.day09;

/*
 * CONSTRUCTOR
 * SETTER
 */
public class Student {
 String name;
 int id;
 String address;
   Resume resume;

 public Student() {
     System.out.println("Student class constructor called ");
 }

 public Student(String name, int id, String address,Resume resume) {
  super();
  System.out.println("parameterized constructor called");
  this.name = name;
  this.id = id;
  this.address = address;
  this.resume=resume;
 }

 public String getName() {
  return name;
 }

 public void setName(String name) {
  System.out.println("name setter called");
  this.name = name;
 }

 public int getId() {
  return id;
 }

 public void setId(int id) {
  System.out.println("Id setter called");
  this.id = id;
 }

 public String getAddress() {
  return address;
 }

 public void setAddress(String address) {
  System.out.println("ADDRESS SETTER CALLED");
  this.address = address;
 }

 public Resume getResume() {
   return resume;
 }

 public void setResume(Resume resume) {
  System.out.println("Resume SETTER CALLED");
   this.resume=resume;
 }

 @Override
 public String toString() {
  return "Student [name=" + name + ", id=" + id + ", address=" + address + ", resume=" + resume + "]";
 }


}
```

```java
package com.day09;

public class Resume {
  public String degree;
  public int experience;
  public Resume() {
    System.out.println("Resume class constructor called");

  }
public Resume(String degree, int experience) {
 super();
 System.out.println("Resume class parameterized");
 this.degree = degree;
 this.experience = experience;
}
public String getDegree() {
 return degree;
}
public void setDegree(String degree) {
 System.out.println("Resume degree setter called");
 this.degree = degree;
}
public int getExperience() {
 return experience;
}
public void setExperience(int experience) {
 System.out.println("Resume experience setter called");
 this.experience = experience;
}
@Override
public String toString() {
 return "Resume [degree=" + degree + ", experience=" + experience + "]";
}


}
```

```xml
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
 xmlns:context="http://www.springframework.org/schema/context"
 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

 xsi:schemaLocation="
        http://www.springframework.org/schema/beans
        https://www.springframework.org/schema/beans/spring-beans.xsd
        http://www.springframework.org/schema/context
        https://www.springframework.org/schema/context/spring-context.xsd">

 <bean id="studentResume" class="com.day09.Resume">
  <!--  <property name="degree" value="B.tech"></property>
  <property name="experience" value="1"></property> -->

  <constructor-arg value="B.tech"></constructor-arg>
  <constructor-arg value="1"></constructor-arg>
 </bean>

 <bean id="studentData" class="com.day09.Student">
  <!--  <property name="name" value="prashant"></property>
  <property name="id" value="101"></property>
  <property name="address" value="hyd"></property>
  <property name="resume" ref="studentResume"></property> -->

   <constructor-arg value="prashant"></constructor-arg>
   <constructor-arg value="101"></constructor-arg>
   <constructor-arg value="hyd"></constructor-arg>
   <constructor-arg ref="studentResume"></constructor-arg>


 </bean>



</beans>
```

```java
public class App {
 public static void main(String[] args) {

  /*   Traditional approach
   * System.out.println("\u001B[31m"+"with java approach");
   * Resume resume=new Resume();
   * resume.setDegree("b.tech");
   * resume.setExperience(0);
   *
   * Student student1=new Student();
   * student1.setId(101);
   * student1.setAddress("hyd");
   * student1.setName("prashant");
   * student1.setResume(resume);
   *
   * System.out.println(student1);
   */

  System.out.println("\u001B[32m" + "with spring approach" + "\u001B[0m");
  ApplicationContext context = new ClassPathXmlApplicationContext("com/day09/application-context.xml");
  /*
   * If the value is injected by the IOC container through setter then that
   * approach is known as setter Injection
   *
   * setter-> property tag
   *
   * and setter will only work when the class object is created
   *
   *
   * If the value is Injected by the IOC container through Constructor then that
   * approach is known as constructor injection
   *
   * Constructor -> constructor-arg
   *
   * constuctor will execute when IOC CONTAINER CREATE
   *
   */
  Student bean = context.getBean("studentData", Student.class);
  System.out.println(bean);

 }
}
```